

# Dynamic Response of Antiresonant Hollow-Core Fibre Drawing to Pressure Perturbations

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**Abstract** *A transient thermo-viscous model for drawing antiresonant hollow-core fibre is presented and experimentally validated. Core and capillary pressure perturbations drive distinct geometric responses, with inter-capillary gap showing the largest sensitivity. Low-frequency disturbances induce the strongest modulation, providing guidance for stable pressure control during fabrication. ©2026 The Author(s)*

## Introduction

Antiresonant hollow-core fibres (AR-HCFs) have emerged as a promising platform for low-loss and low-nonlinearity light transmission because optical power is guided predominantly in air rather than in silica [1, 2]. Owing to these properties, they are attractive for applications such as optical communications and advanced photonic delivery systems [2-4]. Their performance, however, is strongly dependent on structural precision: relatively small deviations in core size, capillary dimensions, wall thickness, or inter-capillary spacing can noticeably affect confinement loss, transmission bandwidth, and modal behaviour [1, 2]. The fabrication of AR-HCFs is based on a pressure-assisted draw process in which viscous deformation, surface tension, axial stretching, and internal gas pressure interact to determine the final microstructure [1, 5, 6]. Earlier modelling studies established the basic mechanics of capillary and microstructured-fibre drawing and clarified the role of pressure in shaping internal holes during the neck-down process [7]. More specifically, Jasion *et al.* [1] analysed AR-HCF drawing and highlighted important fabrication constraints, including the risk of mid-draw contact (MDC) between neighbouring capillaries. Transient and stability studies on hollow or microstructured fibre drawing have shown that time-dependent perturbations can influence the final geometry and may lead to resonance-like or sensitivity effects [7-11]. Recent work has highlighted the value of real-time monitoring and feedback control in hollow-core fibre fabrication [12]. However, the transient response of AR-HCF drawing to time-varying pressure perturbations remains poorly understood, especially when disturbances are applied separately to the core and capillary regions. Since pressure is a key control parameter during draw, its dynamic variation can modify the fibre geometry through different mechanisms and potentially increase the risk of structural defects.

This paper investigates the transient dynamics of tubular (e.g. with circular, non-nested

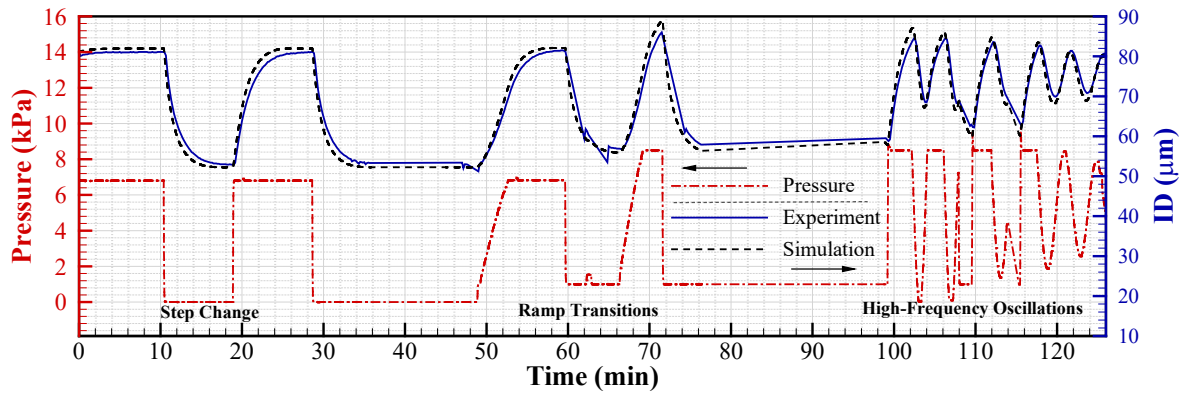
capillaries) AR-HCF drawing under time-dependent perturbations in core and capillary pressures. A coupled thermo-viscous model is employed to analyse the evolution of key geometric parameters, and the transient framework is validated using measurements from a pressurised capillary drawing experiment. The results show that the drawing process behaves as a low-pass dynamic system, that core and capillary pressure disturbances act through different deformation pathways, and that the inter-capillary gap is the most sensitive geometric parameter. These findings provide practical guidance for pressure control in the fabrication of antiresonant hollow-core fibres.

## Transient Model of Pressure-Controlled AR-HCF Drawing

Tubular AR-HCF drawing was modelled using the time-dependent form of the established slender-body thermo-viscous framework described in [1, 6]. Unlike the steady-state formulation, the model resolves the structure in axial position and time, enabling analysis of time-varying pressure perturbations. The molten silica was modelled as an incompressible Newtonian fluid in an axisymmetric geometry with an axially varying temperature field. The model captures the nonlinear coupled action of viscous flow, surface tension, axial stretching, and independently imposed core and capillary pressures on the evolving jacket and capillary dimensions. The resulting six coupled time-dependent nonlinear equations were solved numerically using a finite-element formulation with implicit time integration.

## Experimental Validation

The transient model was validated against a pressurised single-capillary drawing experiment, as shown in Fig. 1, which compares the applied pressure input with the measured and simulated inner-diameter responses. A silica capillary tube with an initial outer/inner diameter (OD/ID) of 20/5 mm was drawn to a final fibre OD of 211  $\mu\text{m}$ .



**Fig. 1:** Experimental validation of the transient model. Applied pressure and measured inner-diameter response are compared with the simulated dynamics.

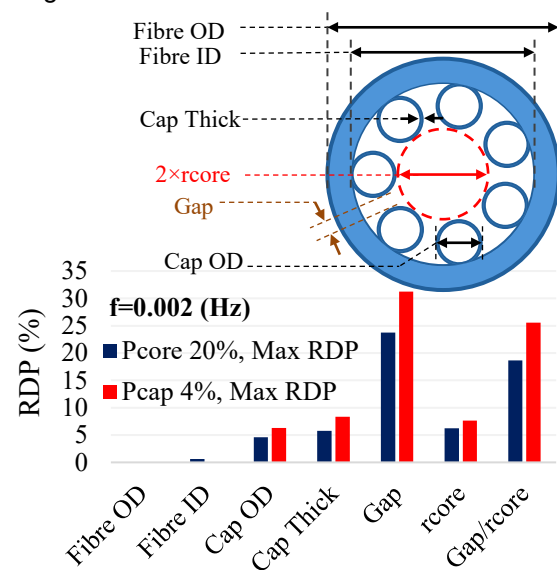
The internal pressure was varied in stepped, ramped, and oscillatory ways, while the fibre inner diameter was measured non-destructively in real time [12]. In the simulation, the measured pressure history was applied directly as a time-dependent boundary condition without fitting parameters. The predicted inner-diameter response agreed well with the experimental data across the full pressure programme, reproducing both transient and oscillatory behaviour, and confirming that the model can capture the main dynamics of pressure-controlled deformation during draw.

#### Dynamic Response to Pressure Perturbations

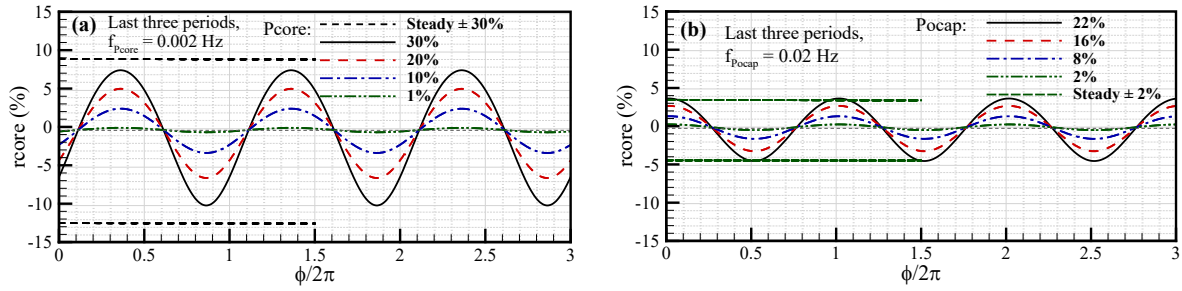
The reference structure, shown in Fig. 2, is a 7-tube AR-HCF preform with seven identical capillaries surrounding a hollow core. It has an outer diameter of  $\sim 10$  mm and is drawn to a fibre diameter of  $\sim 125$   $\mu\text{m}$  under a tension of  $\sim 200$  g. To analyse oscillatory forcing, the core or capillary pressure was prescribed as  $P(t) = P_0(1 + A \sin(2\pi ft))$ , where  $P_0$  is the steady pressure,  $A$  is the relative perturbation amplitude expressed as a fraction of  $P_0$  and corresponding to the excitation percentages given in the figure legends, and  $f$  is the excitation frequency. The outer and inner radii of the jacket tube and capillaries were solved as functions of axial position and time, with inlet conditions specified by the preform geometry, feed speed, and furnace temperature profile. All transient geometric responses were reported relative to the steady-state solution using the relative deviation percentage, defined as  $\text{RDP}(\%) = 100 \times (X - X_{ss})/X_{ss}$ , where  $X$  denotes the instantaneous geometric variable and  $X_{ss}$  its steady-state value at  $t=0$ . This normalisation enables direct comparison of the dynamic sensitivity of parameters such as core radius, capillary thickness, and inter-capillary gap under different pressure perturbations.

The 7-tube AR-HCF was first drawn under steady-state conditions, after which the transient response to the core and capillary pressures

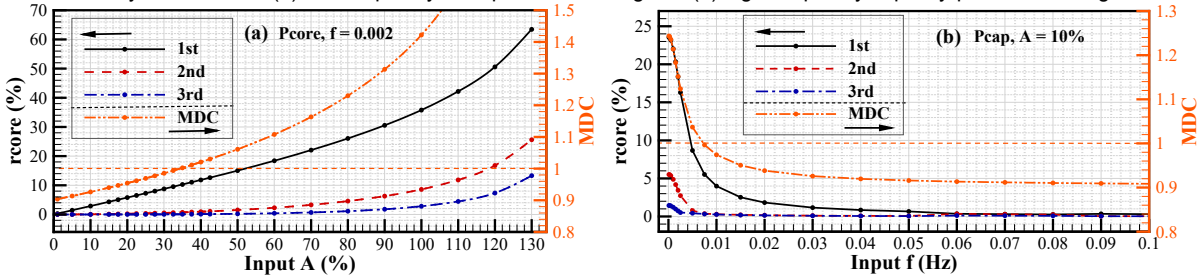
perturbations was computed over several hundred forcing periods and the periodic structural response was analysed after the initial transients had decayed. As shown in Fig. 2, for a slow perturbation (0.002 Hz) the inter-capillary gap exhibits the largest absolute periodic deviation among the main geometric parameters, while the fibre outer diameter changes only weakly; the absolute maximum periodic RDP denotes the largest magnitude of the relative deviation over one forcing cycle after the initial transients have decayed. This identifies the gap as the most sensitive structural quantity under oscillatory pressure forcing. The dynamic behaviour is illustrated in Fig. 3(a,b) through the two representative cases with different excitation conditions. Here, 0.002 Hz and 0.02 Hz correspond to periods of 500 s and 50 s, respectively. These plots illustrate the contrast between larger and smaller periodic responses, while the dependence on amplitude and frequency is examined more systematically in Fig. 4.



**Fig. 2:** AR-HCF cross-section and absolute maximum periodic RDP for the main geometric parameters under low-frequency core and capillary pressure perturbations.



**Fig. 3:** Core-radius response under harmonic pressure excitation, with horizontal dashed lines indicating the corresponding steady-state values: (a) low-frequency core-pressure forcing and (b) high-frequency capillary-pressure forcing.



**Fig. 4:** First, second, and third harmonic core-radius response and MDC parameter, which quantifies the proximity of capillaries to contact and reaches unity at contact, plotted against (a) core-pressure amplitude and (b) capillary-pressure frequency.

Different excitation amplitudes were used because core and capillary pressure perturbations lead to different sensitivities, and the selected values therefore provide clear representative comparisons. Under low-frequency *core*-pressure excitation, the core radius exhibits a large and nearly sinusoidal variation, indicating that the structure follows the forcing in a quasi-static manner which is confirmed by comparing with the steady response shown as horizontal dashed lines. By contrast, the representative high-frequency *capillary*-pressure excitation produces a much smaller modulation, showing that the molten glass cannot fully respond to rapid forcing. This behaviour is consistent with low-pass dynamic response. Core-pressure perturbations mainly modify the global radial stress balance and therefore affect both jacket and internal dimensions, whereas capillary-pressure perturbations act more locally on the thin antiresonant membranes.

Fig. 4(a,b) shows how the harmonic core-radius response depends on perturbation amplitude and frequency. The second and third harmonic indicate nonlinearity, since a linear response contains only the first harmonic. In Fig. 4(a), increasing the core-pressure amplitude produces a strong rise in the harmonic core-radius modulation because larger pressure excursions generate stronger radial deformation of the molten structure. At the same time, the normalised MDC parameter, which measures the proximity of neighbouring capillaries to contact, increases, indicating that neighbouring capillaries move closer to contact;  $MDC < 1$  indicates no contact, while

$MDC \geq 1$  corresponds to capillary contact, which is undesirable and irreversible [1]. In Fig. 4(b), increasing excitation frequency reduces both the response amplitude and MDC. This occurs because the viscous glass cannot fully follow rapid pressure oscillations during its finite residence time in the hot zone, so the deformation is dynamically attenuated. As a result, large low-frequency perturbations pose the greatest risk of mid-draw contact, whereas high-frequency forcing is progressively suppressed.

## Conclusions

A transient thermo-viscous model has been developed for pressure-controlled drawing of tubular AR-HCFs and validated by pressurised capillary-draw experiments. Applied to a 7 tube AR-HCF, the results show that the draw process behaves as a low-pass dynamic system, with larger geometric modulation at low-frequencies and attenuated response at higher frequencies. Core- and capillary-pressure perturbations act through different deformation pathways, while the inter-capillary gap is identified as the most sensitive geometric parameter. These findings provide practical guidance for pressure-control strategies that improve geometric stability during AR-HCF fabrication to give high-yield low loss fibres. With further development, the model could be extended to more complex hollow-core designs, such as ultra-low-loss DNANFs, and could help improve understanding of draw dynamics to enhance longitudinal consistency and suppress unwanted resonant features during fabrication.

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